

Towards Rare Social Event Prediction via Mediator Learning

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Abstract

Rare social events are infrequent yet influential incidents. Predicting such events is practically significant yet inherently challenging due to their extreme scarcity in web-based event stream. Existing studies view this task as an imbalanced classification problem and adopt static rebalancing methods to mitigate scarcity. However, they (1) ignore inter-event dependency that represents the interactions between different event streams, failing to capture precursors that lead to rare events and fundamentally limits performance. They (2) overlook intra-event dependency between different time points within single event stream, which prevents the model from adapting to shifting event patterns and degrades its generalization ability. To this end, we propose a novel **Mediator Learning (ML)** framework, which introduces mediators to explicitly model complex dependencies within web-based event streams. Specifically, we propose (1) Precursor Event Router (PER) that utilizes an information-theoretic routing approach to extract precursor events as mediators from massive event streams. Based on extracted mediators, (2) Conditional Hierarchical Graph Network (CHG) is introduced to model observed events, mediators and rare events into bottom-up graph levels, where its upper-level propagation is conditioned on bottom-level probability distribution. Jointly, these two modules decompose the imbalanced task into two more balanced stages, which not only mitigates the scarcity of rare events but also explicitly model inter-event dependencies, so as to capture precursor events leading to target rare events. Finally, we design (3) Adaptive Information Regularizer (AIR) to optimize the two stages. It dynamically adjusts the information flow between two stages, which models intra-event dependencies and facilitate adaption to drifting event patterns. We theoretically reveal the effectiveness of *ML* by framing it within Information Bottleneck (IB) principle. Extensive experiments show our superior accuracy and interpretability compared with SOTA.

CCS Concepts

• Information systems → Web mining.

Keywords

Social event; time series classification; information bottleneck

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WWW '26, Dubai, United Arab Emirates.
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ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792234>

ACM Reference Format:

Mingjie Qiu, Zhiyi Tan, and Bing-Kun Bao. 2026. Towards Rare Social Event Prediction via Mediator Learning. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792234>

1 Introduction

Social events are collective incidents that have public impact. In the information ecosystem of the Web, these events are usually recorded in massive web-based data stream as news reports [35] and social media posts [29], providing a view for observation. Rare social events (e.g. conflicts and chaos) are a unique subset of social events due to their major impacts. Predicting such event is significant for computational social science, but also inherently challenging due to their extreme scarcity within the event stream. Liu et al. [22] formulate this as "Curse of Rarity", which describes a fundamental dilemma that extreme scarcity leads to limited data points, which further hinders reliable prediction.

To address these challenges, existing studies typically view rare social event prediction task as an imbalanced classification problem. They can be categorized into two groups: (1) data-level rebalancing approaches such as oversampling [14, 37], undersampling [23, 34, 42] and data augment [27, 39], and (2) algorithm-level rebalancing approaches such as cost-sensitive learning [12, 13, 21], contrastive learning [11, 18, 19] and disentangled learning [40]. However, these methods share two drawbacks: (1) they treat rare events isolatedly and **ignore inter-event dependencies**. The web-based event data usually contains rich interactions across different event streams [24, 26]. These interactions serve as crucial precursor signals, which are ignored by the rebalancing strategies of existing methods, and (2) they **face challenges in modeling intra-event dependencies**. Even within a single event stream, there often exists significant temporal non-stationarity (i.e., highly uneven temporal distribution), which requires the model to consider the dependency among time points and dynamically adapt to the drifting event patterns. Therefore, relying solely on static rebalancing methods may introduce outdated correlations, resulting in poor generalization and skewed feature space, as shown in Figure 1(a).

To this end, we introduce precursor events as mediators to bridge observed events and rare events. Specifically, we propose a novel **Mediator Learning (ML)** framework, which is composed of three modules: Firstly, we propose Precursor Event Router (PER), which measures the transition probability between different social events and performs information-theoretic routing. This extracts critical precursor events as mediators from massive web-based event streams. Secondly, Conditional Hierarchical Graph Network (CHG)

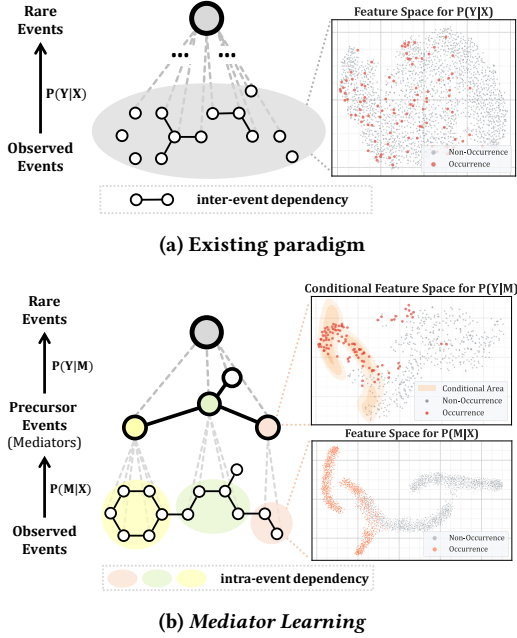


Figure 1: Illustration of our motivations: (a) existing paradigm suffers from extreme scarcity of rare social events in event streams, (b) our paradigm introduces mediators to decompose the non-trivial problem into two balanced stages.

is proposed to model observed events, precursor events and rare events into a bottom-up hierarchical graph, where upper-level information propagation is conditioned on probability distribution of bottom-level precursor events. These two modules jointly **decompose the non-trivial imbalanced task into two balanced stages**: (1) Extracting precursor events from observed events, which alleviates scarcity issue by focusing on frequent precursor events. (2) Predicting rare events based on precursor events. This models the interactions between different event streams and enables the learning of conditional feature space, which improves discriminative power and predictive performance, as shown in Figure 1(b). Lastly, we develop an Adaptive Information Regularizer (AIR) to model intra-event dependencies and supervise the learning of aforementioned two stages. It enables the model to **dynamically align its internal information flow with shifting event patterns**, so as to handle non-stationarity within each single event stream. Moreover, AIR adopts both compressive and predictive constraints, which **enables direct supervision and adaptive balancing of two stages**, so as to improve generalization and robustness.

To demonstrate the effectiveness of our proposed ML framework both theoretically and empirically, we first prove that Information Bottleneck (IB) principle can be formulated as the learning objective of ML. Moreover, extensive experiments on real-world datasets demonstrate that ML provide superior performance and interpretable insights. Our contributions can be summarized as:

- We propose a novel paradigm called *Mediator Learning*, which contributes both methodologically and theoretically to the community of rare social event prediction.

- We design a comprehensive model to implement this paradigm, featuring PER for mediator extraction, CHG for mediator modeling and inter-event dependency, and AIR for two-stage joint optimization and intra-event dependency.
- We theoretically demonstrate the validity of *Mediator Learning* from the Information Bottleneck (IB) principle.
- We conduct experiments on real-world datasets, validating our competitive performance and insightful interpretations.

2 Related Works

2.1 Event Prediction

In previous study, rare event prediction is usually viewed as a binary Time Series Classification (TSC) task [33, 36], whose goal is to predict the occurring probability based on historical event time series [2, 8]. For instance, ROCKET [6] transforms time series using random convolutional kernels and combines them with linear classifier to achieve competitive classification accuracy. PatchTST [28] reduces computational complexity by segmenting time series into patches as input tokens. Series2Vec [10] uses the prediction of similarity between pairs of sequences from the same class as a self-supervised task. Moreover, TSCMamba [1] utilize multi-view learning to assist the TSC task. However, these methods are primarily designed for balanced cases and seldom consider class imbalanced scenarios, which limits their practical applications on rare event prediction.

2.2 Imbalanced Learning

Imbalanced Learning aims to learn on data with highly imbalanced class distribution [16]. Existing methods mainly focus on data-level [14, 34] and algorithm-level rebalancing methods [18]. Data-level methods can be broadly categorized into three strategies: oversampling, undersampling, and hybrid sampling. For example, SMOTE [3] synthesizes new samples by linearly interpolating between minority class samples and their neighbors in the feature space. T-SMOTE [41] further extends SMOTE to the generation of minority class samples in time-series scenarios. Algorithm-level methods focus more on the minority class. Focal Loss [21] modifies the standard cross-entropy loss to down-weight well-classified examples, thus focusing training on minority samples. SmdCLR [19] proposes a semi-supervised framework to address class imbalance via graph convolutional projection and semantic regularization. However, manipulating data distribution or learning process can easily introduce fake patterns, leading to performance degradation.

2.3 Information Bottleneck

Information Bottleneck theory (IB) aims to find an optimal data representation that maximally filters noise and preserves useful information [4, 17, 32]. TimeCIB [4] apply conditional information bottleneck for imputation, so as to reduce redundant information while preserving temporal contexts. InsGNN [9] leverages IB to achieve interpretability in both temporal and spatial dimensions by extracting invariant causal subgraphs. Existing methods often model IB under static conditions, which limits its effective application in dynamic social contexts. However, these studies reveal a critical gap when predicting rare events. Existing methods typically condition the bottleneck on an explicit and relatively static context,

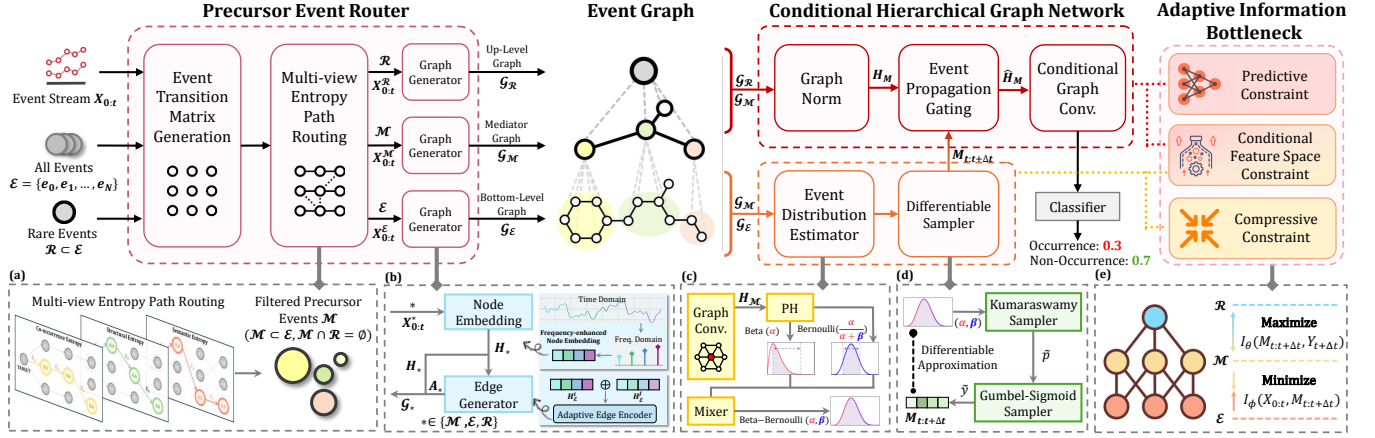


Figure 2: Workflow of *Mediator Learning*, a novel rare event prediction paradigm containing a Precursor Event Router (PER), a Conditional Hierarchical Graph Network (CHG) and an Adaptive Information Bottleneck (AIR).

which is effective for tasks like time series imputation [4], whose goal is to preserve local dependencies. In rare event prediction, the underlying event patterns shift over time, which requires a more dynamic and adaptive compression strategy. Our work addresses this limitation by proposing a novel adaptive mechanism where the bottleneck is conditioned not on raw contextual data, but on a dynamically learned representation of the evolving social contexts, enabling the model to actively adapt to non-stationarity.

3 Preliminary

3.1 Rare Events and Precursor Events

Following previous works [42], we define rare social events by their occurrence rate λ , which represents the number of occurrence per unit time. A lower λ indicates that the event is rarer. To keep accordance with previous works [36], we categorize rare events into three distinct levels: R1-level extremely rare events with λ among 0–1%, R2-level very rare events with λ among 1–5%, and R3-level moderately rare events with λ among 5–10%.

In our work, a precursor event must meet the following core criteria: (1) It must occur before the target event; (2) A clear predictive association with target event; (3) Occurrence rate must satisfy $\lambda > 10\%$. Notably, the precursor events function as predictive mediators rather than causal ones, since the real-world environment makes it difficult to consider all confounding factors. Therefore, our goal is not to estimate intervention effects, but to boost predictive performance via observed evolutionary paths between events.

3.2 Problem Formulation

The task of social event prediction is often considered as a time series classification problem. Assuming we have a universal set $\mathcal{E} = \{e_1, e_2, \dots, e_{|\mathcal{E}|}\}$ containing $|\mathcal{E}|$ different types of social events, then historical events can be represented as event stream $X_{0:t}$, which represents occurrence number at each time point. The objective is to predict whether target rare event Y will occur in a future prediction window Δt . Therefore, their core task is to learn a mapping function that estimates the probability of rare event $P(Y_{t+\Delta t}|X_{0:t})$. However,

it is non-trivial to learn a direct mapping function due to extreme class imbalance problem. In contrast, our paradigm aims to find a subset of precursor events $\mathcal{M} \subset \mathcal{E}$ as mediators:

$$P(Y_{t+\Delta t}|X_{0:t}) = \sum_{\mathcal{M}} P(M_{t:t+\Delta t}|X_{0:t}) \cdot P(Y_{t+\Delta t}|M_{t:t+\Delta t}) \quad (1)$$

where $M_{t:t+\Delta t}$ refers to the probability distribution of mediator occurrences among t and $t + \Delta t$. As the computational cost of Eq.(1) is expensive, we introduce a more tractable and learnable information-theoretic objective as an alternative:

Lemma 1. Information Bottleneck (IB) Principle. A probabilistic process shown in Eq.(1) can be approximated by optimizing a surrogate information bottleneck Lagrangian:

$$\min_{\phi, \theta} I_{\phi}(X_{0:t}; M_{t:t+\Delta t}) - \gamma I_{\theta}(M_{t:t+\Delta t}; Y_{t+\Delta t})$$

where $I(\cdot; \cdot)$ denotes the mutual information, γ controls the trade-off between compression and preservation, and ϕ, θ correspond to learnable parameters that define probabilistic mappings.

4 Methods

In this section, we present the detailed workflow of *Mediator Learning* as shown in Figure 2. Notations are available in **Appendix A**.

4.1 Precursor Event Router

The Precursor Event Router (PER) module is designed to identify precursor events $\mathcal{M}$ from event stream $\mathcal{E}$. This module contains an event transition matrix generation block, a multi-view entropy path routing block and three graph generator blocks.

4.1.1 Event Transition Matrix Generation. These matrix measures the transition probability of one event transform into another within the event streams $\mathcal{E}$. We utilize three kinds of scores to measure the transition probability: (1) Structural score T_h encodes the structural likelihood to measure hierarchical relations via CAMEO coding scheme (e.g. "02" and "02.1"). (2) Co-occurrence score T_o measures the likelihood of two events co-occurring within a given

time window. (3) Semantic score T_s encodes the semantic likelihood by measuring the similarity of descriptive texts. A higher similarity indicates a higher likelihood of transformation (e.g. "Appeal for judicial cooperation" and "Reject judicial cooperation"). The transition probability T from event e_i to e_j can be expressed as:

$$\begin{aligned} T_h &= \text{dist}(e_i, e_j), \quad T_s = f_\Theta(\mathbf{v}_i, \mathbf{v}_j), \quad T_o = H(I, J) \\ T_{ij} &= ([T_h, T_o, T_s] + \epsilon)^{-1} \in \mathbb{R}^3 \end{aligned} \quad (2)$$

where $\text{dist}(\cdot, \cdot)$ refers to the shortest structural distance, $f_\Theta(\cdot, \cdot)$ is a semantic similarity function, $\mathbf{v}$ is the semantic embedding of the descriptive texts for events, $H(\cdot, \cdot)$ is the joint entropy for random variables I and J representing the occurrence of event e_i and e_j , ϵ is a computational bias. The obtained edge vectors $T_{ij} \in \mathbb{R}^3$ further form the event transition matrix $\mathbf{T} \in \mathbb{R}^{|\mathcal{E}| \times |\mathcal{E}| \times 3}$. More discussions on event transition matrix can be found in **Appendix D**.

4.1.2 Multi-view Entropy Path Routing. Based on the obtained event transition matrix $\mathbf{T}$, we further propose a multi-view path routing module to identify precursor events that leading to the target rare events. Specifically, our goal is to find high-probability paths within transition matrix $\mathbf{T}$. However, in conventional probability frameworks, path probability is defined as a continuous product of transition probabilities along the pathway. This leads to numerical underflow issues, which interfere with the training process. To this end, we introduce information entropy as a reliable proxy, which transfer the continuous product of probability to cumulative sum of entropy. We utilize negative log-likelihood to transfer the path probability into entropy:

$$\log(P(\Pi)) = \sum_{(i,j) \in \Pi} \log(T_{ij}) \quad (3)$$

where Π refers to all probability paths leading to rare event $\mathcal{R}$. Next, we apply Soft-Dijkstra searching method [25] on the transition matrix $\mathbf{T}$, which enables end-to-end differentiability and includes the PER module into optimization process. In this way, We find high-probability paths and consider the nodes on these paths to be precursor events $\mathcal{M}$.

4.1.3 Graph Generator. Based on the set of rare events $\mathcal{R}$, precursor events $\mathcal{M}$, observed events $\mathcal{E}$ and their event streams $X_{0:t}^{\mathcal{R}}, X_{0:t}^{\mathcal{M}}, X_{0:t}^{\mathcal{E}}$, we construct the spatio-temporal graph as shown in Figure 2(b). We first utilize a node embedding layer to extract temporal features on time series and form the node embeddings for each event. We transform a time-domain time series into trend and seasonality components. Following the practice of N-Beats [30], we utilize low-order polynomials and Fourier series as basis functions to extract features from the time-domain time series:

$$H_* = \text{NodeEmb}(X_{0:t}^*) \quad * \in \{\mathcal{R}, \mathcal{M}, \mathcal{E}\} \quad (4)$$

where H_* refers to the obtained node embeddings for each event type. Furthermore, we generate the adaptive edge for each node pair. The edge weight from event e_i to e_j can be expressed as:

$$A_*^{ij} = \text{Relu}(\rho^T \cdot \text{Concat}(H_*^i, I_i, H_*^j, I_j)) \quad * \in \{\mathcal{R}, \mathcal{M}, \mathcal{E}\} \quad (5)$$

where I_i and I_j are the identity embeddings of node i and j , ρ refers to a trainable matrix. In this way, we form the adjacent matrix $\mathbf{A}_*$ and forms the graph for observed events $\mathcal{G}_{\mathcal{E}} = \{H_{\mathcal{E}}, \mathbf{A}_{\mathcal{E}}\}$, precursor

events $\mathcal{G}_{\mathcal{M}} = \{H_{\mathcal{M}}, \mathbf{A}_{\mathcal{M}}\}$ and rare events $\mathcal{G}_{\mathcal{R}} = \{H_{\mathcal{R}}, \mathbf{A}_{\mathcal{R}}\}$. These graphs measures both the temporal and relational interactions between events, laying the foundations for following steps.

4.2 Conditional Hierarchical Graph Network

The Conditional Hierarchical Graph Network module is implemented via a two-stage, bottom-up conditional information propagation within a hierarchical graph structure. Specifically, this heterogeneous graph encodes all observed events $\mathcal{E}$, precursor events $\mathcal{M}$ and target rare events $\mathcal{R}$ into bottom-up graph levels. Within the cross-level information propagation, instead of directly passing prediction result, CHG allows the bottom-level network to pass a probability distribution $M_{t:t+\Delta t}$ to upper-level network. This introduces uncertainties and mitigates propagation errors.

4.2.1 Event Distribution Estimator. Firstly, we estimate the probability distribution $M_{t:t+\Delta t}$ for mediators $\mathcal{M}$ via graph $\mathcal{G}_{\mathcal{E}}$. We first utilize a spatio-temporal encoder to encode input $X_{0:t}$ into mediator embeddings $\mathbf{E}_m$, as shown in Figure 2(c). We first apply graph convolutions on $\mathcal{G}_{\mathcal{E}} = \{H_{\mathcal{E}}, \mathbf{A}_{\mathcal{E}}\}$. Technically, any graph convolution method can be applied here. In our practice, we utilize the native GCN for aggregation:

$$\mathbf{E}_m = f_{\text{GCN}}(H_{\mathcal{E}}, \mathbf{A}_{\mathcal{E}}) \quad (6)$$

Next, we use a parameter head (denoted as PH in Figure 2(c)) to estimate two critical parameters for Beta-Bernoulli distribution. The input-output schema and its PMF are:

$$\begin{aligned} (\alpha, \beta) &= \text{Softplus}[f_{\text{MLP}}(\mathbf{E}_m)], \quad \omega \sim \text{Beta}(\alpha, \beta) \\ p_{\text{PMF}}(x, \omega | \alpha, \beta) &= \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \omega^{\alpha-1} (1-\omega)^{\beta-1} \end{aligned} \quad (7)$$

where $\mathbf{E}_m$ is the aggregated representations of mediators, Softplus is the smooth function, (α, β) denote the two shape parameters of the Beta-Bernoulli distribution. Γ is the Gamma function. ω is the estimated probability of event occurrence. The reason we mix the beta distribution and Bernoulli distribution is that it not only provides the probability, but also output the probability confidence, which is significant for optimization. More explanations on Beta-Bernoulli distribution could be checked in **Appendix E**.

4.2.2 Differentiable Sampler. In order to obtain conditional information gating, we design a sampling procedure to not only draw samples from the learned continuous probabilities, but also keep the end-to-end differentiability for the workflow. For each precursor event $e_i \in \mathcal{M}$, we perform the two-step sampling procedure.

$$p_i \sim \text{Beta}(\alpha_i, \beta_i), \quad M_{t:t+\Delta t}^i \sim \text{Bernoulli}(p_i) \quad (8)$$

where the first step of probability sampling is key to capturing uncertainty, and the sampled probability value p_i is then used as the parameter of a Bernoulli distribution to sample whether the precursor event occurs. Since directly applying random sampling prevents gradients from back propagating, we use the Kumaraswamy distribution [20] and the Gumbel-Sigmoid sampling as reparameterization techniques. In this way, we obtain $M_{t:t+\Delta t}^i \in (0, 1)$ indicating whether the i -th precursor event occurs. The sampled values of all precursor events together form conditional gating vector $M_{t:t+\Delta t}$.

4.2.3 Conditional Graph Convolution. In this module, we utilize the sample gating vector $M_{t:t+\Delta t}$ to enable conditional upper-level information propagation:

$$\hat{\mathbf{E}}_m = M_{t:t+\Delta t} \odot \mathbf{E}_m \quad (9)$$

where $\odot$ denotes the Hadamard product, $\hat{\mathbf{E}}_m$ is the learned conditional feature, $M_{t:t+\Delta t}$ is broadcast to match the feature dimension of $\mathbf{E}_m$. This process conditions the upper-level information aggregations on the bottom-level probability distributions. Next, we utilize a GCN for upper-level information propagation and a simple MLP classifier for final classification:

$$\mathbf{E}_r = f_{\text{GCN}}(\hat{\mathbf{E}}_m, A_M), \quad y_{\text{pred}} = f_{\text{MLP}}(\mathbf{E}_r) \in (0, 1) \quad (10)$$

where y_{pred} is the final binary classification output.

4.3 Adaptive Information Regularizer

To balance the two learning stages and further optimize the mediator, we further design an Adaptive Information Regularizer (AIR) as shown in Figure 2(e). Considering that the evolution of rare social events exhibits temporal non-stationarity, a fixed, globally shared information compression strategy is more likely to be suboptimal. To this end, we condition the information bottleneck framework on the representation of mediators. The representation $\mathbf{E}_m$ contains significant temporal dynamics that reflect the social contexts, which informs the adaptive adjustment of bottleneck behaviors. Accordingly, we extend the standard information bottleneck in *Lemma 1* to the following adaptive variant:

Lemma 2. Conditional Information Bottleneck Approximation. Derived from *Lemma 1*, the probabilistic process shown in Eq.(1) can be further approximated by an adaptively conditioned information bottleneck Lagrangian:

$$\min_{\phi, \theta} \underbrace{I_\phi(X_{0:t}; M_{t:t+\Delta t}, \mathbf{E}_m)}_{(a)} - \underbrace{I_\phi(X_{0:t}; \mathbf{E}_m)}_{(b)} - \gamma \underbrace{I_\theta(M_{t:t+\Delta t}; Y_{t+\Delta t})}_{(c)}$$

Here, (a)-(b) term is the mutual information conditioned on $\mathbf{H}_m$ and enables adaptive compression, which serves as the conditional feature space constraint. Meanwhile, the (b) term and (c) term serve as the compressive and predictive constraints shown in Figure 2(e). Note that we only apply conditions on bottom-level propagation to follow the optimal compression principle (See **Appendix F**). Our optimization objective $\mathcal{L}_{\text{AIR}}$ then becomes:

$$\begin{aligned} \mathcal{L}_{\text{AIR}} = & \underbrace{\zeta_1 \cdot \mathbb{E}_{p(X, \mathbf{E}_m)} [\text{KL}(p(M_{t:t+\Delta t} | X_{0:t}, \mathbf{E}_m) || p(M_{t:t+\Delta t} | \mathbf{E}_m))]}_{\text{Upper bound of (a)-(b)}} + \\ & \underbrace{\zeta_2 \cdot \mathbb{E}_{p(X)} [||X_{0:t} - \hat{X}_{0:t}||^2]}_{\text{Lower bound of (b)}} - \underbrace{\zeta_3 \cdot \mathbb{E}_{p(M)} [\log p(Y_{t+\Delta t} | M_{t:t+\Delta t})]}_{\text{Lower bound of (c)}} \end{aligned}$$

Here, the (a)-(b) term is implemented by a KL-divergence. We assume both posterior and prior comply to Gaussian distribution and utilize the closed-form analytical solution for loss computation. The (b) term is implemented by a Mean Square Error (MSE) loss produced by a masked autoencoder. We divide the time series $X_{0:t}$ into multiple patches along the time axis and randomly mask most of them. The learned representation $\mathbf{E}_m$ is required to reconstruct the

Table 1: Detailed statistics of our used GDELT dataset, where $|\mathcal{E}|$ and $|\mathcal{R}|$ refer to the number of observed event type and rare event type, respectively.

Dataset	$ \mathcal{E} $	$ \mathcal{R} $	#Days	#Train	#Valid	#Test
ME	342	75	2,584	25,133,953	3,141,744	2,929,510
EUR	362	68	2,584	38,223,620	4,778,019	4,567,119

whole time series $\hat{X}_{0:t}$. This encourages the GCN to retain sufficient information about $X_{0:t}$ in $\mathbf{E}_m$. The (c) term is implemented by Negative Log-Likelihood, which is equivalent to Binary Cross-Entropy (BCE) Loss. The derivation and further discussion of *Lemma 2* can be checked in **Appendix F**.

4.4 Optimization Target

To optimize *Mediator Learning*, we design two learning objectives to guide the learning process. Aside from $\mathcal{L}_{\text{AIR}}$, we also propose $\mathcal{L}_{\text{PER}}$ to optimize the PER module. To be specific, we score and optimize the quality of the transition path P_{path} generated by PER. We define a differentiable path quality score function and then train PER by maximizing this score (i.e., by minimizing its negative value). The following two score functions are deployed: (1) Cumulative probability function. A reasonable path should consist of a series of high-probability event transitions. Via event transition matrix $\mathbf{T}$, we calculate the sum of the log probabilities for all transitions along the path. (2) Regularization function. We use a regularization term to encourage PER to find more concise paths by penalizing the path length. This prevents the model from selecting lengthy paths that contain a large number of irrelevant events:

$$\mathcal{L}_{\text{PER}} = w_{\text{trans}} \cdot \sum_{(i,j) \in \Pi} \log(T_{ij}) + w_{\text{len}} \cdot \text{Length}(\Pi) \quad (11)$$

where w_{trans} and w_{len} are predefined weights. $\mathcal{L}_{\text{PER}}$ is combined with $\mathcal{L}_{\text{AIR}}$ to form final loss function for end-to-end joint training:

$$\mathcal{L} = \mathcal{L}_{\text{AIR}} + \kappa \mathcal{L}_{\text{PER}} \quad (12)$$

where κ is a hyperparameter to balance two loss functions.

5 Experiments

In this section, we aim to answer the following research questions:

- **RQ1:** How does *Mediator Learning* perform on real-world dataset compared with state-of-the-art methods?
- **RQ2:** How does each individual component contribute to the overall performance?
- **RQ3:** How does our proposed AIR module address the temporal non-stationarity and enhance robustness?
- **RQ4:** How dose the *Mediator Learning* framework enhance interpretability?

5.1 Dataset Details

We conducted our experiments on datasets derived from the Global Database of Events, Language, and Tone (GDELT), a global repository that monitors a vast and diverse range of web-based news

Table 2: The results for EUR dataset (Europe) and ME dataset (Middle East). The best-performing model is marked by boldface, and the sub-optimal model is marked by underline. λ refers to event occurrence rate, a lower λ indicates a rarer event.

EUR		R1 ($0 < \lambda < 1\%$)			R2 ($1\% < \lambda < 5\%$)			R3 ($5\% < \lambda < 10\%$)			Total ($0 < \lambda < 1$)		
		BACC $\uparrow$	F1 $\uparrow$	MCC $\uparrow$	BACC $\uparrow$	F1 $\uparrow$	MCC $\uparrow$	BACC $\uparrow$	F1 $\uparrow$	MCC $\uparrow$	BACC $\uparrow$	F1 $\uparrow$	MCC $\uparrow$
GRU	<i>NeurIPS'14</i>	57.57	47.34	21.97	60.28	48.94	29.92	64.52	50.13	35.54	70.84	69.12	51.53
MiniRocket	<i>KDD'21</i>	61.62	51.01	30.38	66.37	50.69	37.44	70.87	56.71	46.91	74.12	70.07	56.11
MTG	<i>ICDM'22</i>	59.77	50.55	30.08	73.37	56.88	43.15	69.99	50.15	40.60	78.84	78.61	61.15
Series2Vec	<i>DMKD'24</i>	69.37	62.06	<u>43.79</u>	72.35	52.52	48.79	71.53	49.30	42.10	88.18	<u>80.86</u>	<u>78.67</u>
TSCMamba	<i>IF'25</i>	65.10	57.28	37.88	75.32	60.63	55.41	<u>80.87</u>	<u>73.67</u>	58.46	87.67	77.33	76.76
T-SMOTE	<i>IJCAI'22</i>	62.43	51.93	36.66	76.83	62.51	56.25	75.60	69.87	60.60	79.84	77.13	73.43
SmdCLR	<i>TNNLS'25</i>	70.92	59.24	40.71	77.52	67.67	54.13	76.01	67.34	62.43	84.00	81.04	79.53
DGMSCL	<i>NN'25</i>	<u>70.23</u>	<u>63.23</u>	43.23	<u>78.63</u>	<u>69.10</u>	<u>58.22</u>	78.60	70.35	<u>65.23</u>	83.60	82.53	78.29
GIB	<i>NeurIPS'20</i>	64.85	49.01	28.11	67.83	53.51	45.96	72.64	65.53	62.21	77.77	71.24	63.23
InsGNN	<i>IF'25</i>	66.53	61.52	32.23	72.88	66.75	51.69	78.98	70.13	62.17	79.43	75.11	69.00
Ours	<i>This paper</i>	76.81	74.91	52.02	82.28	80.37	68.23	83.96	82.37	71.23	<u>87.88</u>	85.62	80.86
Δ_{sota} (%)		6.58	11.68	8.23	3.65	11.27	10.01	3.09	8.72	6.00	-0.30	4.76	2.19

ME		R1 ($0 < \lambda < 1\%$)			R2 ($1\% < \lambda < 5\%$)			R3 ($5\% < \lambda < 10\%$)			Total ($0 < \lambda < 1$)		
		BACC $\uparrow$	F1 $\uparrow$	MCC $\uparrow$	BACC $\uparrow$	F1 $\uparrow$	MCC $\uparrow$	BACC $\uparrow$	F1 $\uparrow$	MCC $\uparrow$	BACC $\uparrow$	F1 $\uparrow$	MCC $\uparrow$
GRU	<i>NeurIPS'14</i>	57.75	44.60	24.25	60.06	46.78	29.87	62.36	48.65	34.17	70.81	67.54	61.10
MiniRocket	<i>KDD'21</i>	59.33	60.91	39.71	66.76	51.08	38.20	64.16	58.77	50.65	75.55	71.89	66.91
MTG	<i>ICDM'22</i>	52.38	40.22	34.00	66.51	58.46	37.63	40.12	35.43	14.54	74.17	70.12	62.19
Series2Vec	<i>DMKD'24</i>	61.63	60.97	40.88	70.68	<u>68.71</u>	52.92	72.58	<u>73.63</u>	61.47	86.01	81.22	<u>72.21</u>
TSCMamba	<i>IF'25</i>	62.62	58.17	37.38	70.37	59.64	49.60	66.31	60.71	59.23	<u>85.84</u>	<u>82.52</u>	70.52
T-SMOTE	<i>IJCAI'22</i>	65.87	62.30	40.65	69.98	62.31	51.56	70.29	69.23	60.54	80.31	75.60	75.13
SmdCLR	<i>TNNLS'25</i>	65.41	<u>64.70</u>	41.08	71.63	63.37	<u>56.21</u>	74.11	70.22	<u>65.21</u>	81.44	77.24	71.50
DGMSCL	<i>NN'25</i>	<u>66.70</u>	61.11	38.98	<u>72.35</u>	65.65	55.39	<u>75.65</u>	69.93	60.11	80.95	76.93	75.00
GIB	<i>NeurIPS'20</i>	58.74	54.17	<u>41.28</u>	59.30	58.20	39.96	64.52	62.10	52.11	70.56	69.12	65.53
InsGNN	<i>IF'25</i>	61.61	59.94	40.48	63.33	61.41	43.03	68.25	63.21	56.54	73.85	71.98	69.99
Ours	<i>This paper</i>	72.67	71.36	51.10	78.46	75.89	66.71	81.44	81.78	72.00	84.67	85.27	76.51
Δ_{sota} (%)		5.97	6.66	9.82	6.11	7.18	10.50	5.79	8.15	6.79	-1.34	2.75	4.30

sources¹. This dataset contains content in over 100 languages, capturing a wide spectrum of events, cultures, and perspectives directly from the open web. Following the established benchmarks, we extract two regional subsets based on event location: ME (Middle East) and EUR (Europe). We pre-process the web news data into time series for model input. The data spans from February 2015 to March 2022 and is chronologically split into training set (Feb 2015 – Feb 2021), validation set (Mar 2021 – Aug 2021), and testing set (Aug 2021 – Mar 2022) using a ratio of 8:1:1. A detailed statistical breakdown of all used datasets is provided in Table 1.

5.2 Evaluation Protocols

We define prediction tasks based on the frequency of rare event occurrence. Following the classification standards of existing research [36], we divide the target events into three levels: R1 (extremely rare, with an occurrence rate of 0-1%), R2 (very rare, 1-5%), and R3 (moderately rare, 5-10%). Due to the extreme class imbalance, we employ a combination of evaluation metrics better suited to assess model performance on imbalanced datasets: Balanced Accuracy

(BACC), F1-score (F1) and Matthews Correlation Coefficient (MCC). BACC is employed as the main metric. To avoid interfering with the evaluation metrics, we have removed commonly occurring events that always happen. These events consistently occur within observed time range, and therefore have limited value for prediction and would disrupt the assessment. The used baselines are detailed in **Appendix B**. More details on the implementation and metrics can be found in **Appendix C**.

5.3 Main Results (RQ1)

To answer RQ1, we evaluated our model’s performance on two real-world datasets (EUR and ME), with the results summarized in Table 2. We draw several key findings: (1) Our proposed model significantly outperforms existing methods. In prediction tasks for extremely rare (R1), very rare (R2), and moderately rare (R3) events, our model achieves the best or sub-optimal scores on all metrics. This fully demonstrates the effectiveness of the *Mediator Learning* paradigm. (2) The predictive performance increases with the occurrence rate λ . We can see that as the occurrence rate of events increases, all performance metrics show a significant improvement,

¹<https://www.gdeltproject.org/>

Table 3: Ablation Study of the *Mediator Learning* Framework on R1 and Total setting (%) on EUR dataset. Negative values indicate the performance drop of a variant relative to our full model (Ours).

Module	Category	Method / Architecture	R1-BACC	R1-F1	R1-MCC	Total-BACC	Total-F1	Total-MCC
PER	Ours	PER	76.81	74.91	52.02	87.88	85.62	80.86
	Variants	Random routing	-11.51	-13.62	-7.81	-9.35	-10.28	-9.69
		w/o co-occurrence score	-4.13	-3.62	-2.22	-2.65	-3.13	-1.91
		w/o structural score	-2.41	-3.24	-1.11	-0.89	-0.54	-0.25
		w/o semantic score	-3.23	-2.62	-3.14	-1.09	-0.98	-0.34
CHG	Ours	CHG	76.81	74.91	52.02	87.88	85.62	80.86
	Variants	Plain temporal model	-10.81	-9.60	-5.31	-9.81	-8.88	-4.52
		Non-hierarchical graph	-6.51	-6.30	-6.40	-5.37	-4.42	-5.45
		Unconditional graph	-4.67	-5.35	-4.44	-1.57	-1.33	-1.60
AIR	Ours	AIR	76.81	74.91	52.02	87.88	85.62	80.86
	Variants	Standard BCE loss	-7.92	-7.53	-6.06	-5.25	-4.88	-3.20
		Standard IB loss	-3.20	-5.55	-5.48	-0.90	-1.87	-1.44

which aligns with our expectations. This indicates that the rarity of an event determines its prediction difficulty. (3) The advantage of our model is more prominent on F1 and MCC metrics compared to BACC. While BACC primarily measures the model’s average ability to identify positive and negative samples, the F1-score and MCC are more sensitive to False Positives and False Negatives, which is crucial for rare event prediction with its extreme class imbalance. Our model identifies true positive samples more accurately, significantly reducing false alarms (False Positives) and thereby substantially boosting the F1 and MCC scores. (4) It is worth noting that on the BACC metric for the Total setting, our model performs slightly worse than some top-tier TSC methods (e.g., Series2Vec). It can be attributed that in scenarios involving a large number of common events, the model’s advantages in rare event prediction may be averaged out, preventing it from leading on BACC.

Additionally, a cross-comparison across different datasets reveals that the model’s overall performance on the EUR dataset is superior to its performance on the ME dataset. We attribute this phenomenon to the statistical properties: (1) The total number of events in EUR dataset (around 47.6 million) is significantly larger than of ME dataset (around 31.2 million). A larger number of data allows the model to learn richer and more robust event evolutionary process. More samples allow PER module to accurately calculate the transition probabilities between events and identify more reliable paths, thus improving prediction accuracy. (2) Although ME dataset has fewer total events, it contains more types of rare events (75 in ME vs. 68 in EUR). This means the rare events in the ME dataset are not only more sparse but also more diverse. This increases difficulty of learning, making it challenging for models to learn patterns for each rare event types that are as robust as those on EUR dataset.

5.4 Ablation Study (RQ2)

To validate the effectiveness of each component within the framework, we conducted series of ablation studies shown in Table 3.

Firstly, in Preceding Event Router (PER), we find that random routing leads to catastrophic drops (11.51% decrease in R1-BACC).

This demonstrates that randomly selecting precursor events is ineffective and underscores the necessity of our proposed entropy routing methods. Moreover, we remove the three core scores that constitute the event transition matrix. Removing the co-occurrence score results in the most significant performance degradation, which indicates that the event co-occurrence is the strongest signal for identifying precursor events. Removing the structural score and semantic score also lead to noticeable performance drops, proving the importance of multi-view routing, with each perspective providing significant information.

Secondly, we replace the core module in the Conditional Hierarchical Graph (CHG) with a simple time series model (e.g., GRU), a non-hierarchical graph, and an unconditional graph, respectively. The most notable result comes from plain temporal model, which completely removes mediators and graph structure, leading to a significant performance drop. The simplified non-hierarchical graph also results in a significant performance decline, which demonstrates the importance of hierarchically modeling observed events, preceding events, and rare events. Removing conditional gating also leads to a noticeable performance drop, which demonstrates the effectiveness of the gating mechanism. This mechanism better handles uncertainty and prevents errors from being amplified.

Thirdly, to verify the effectiveness of AIR as an optimization objective, we replace it with a standard classification loss and a standard IB loss, respectively. When degraded to standard BCE loss, the model performance drops significantly. This indicates that IB framework itself is crucial for handling noisy event data, as it helps the model learn more robust features by maximizing the compression of irrelevant information. The performance drop caused by standard IB further demonstrates the importance of adaptive design in AIR. By dynamically adjusting compression strategy according to temporal dynamics, AIR can address non-stationarity of rare events more flexibly and achieve better performance.

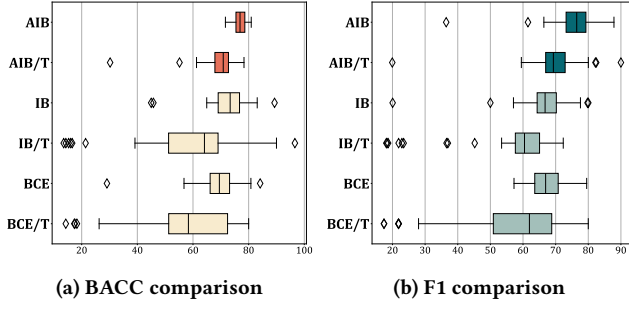


Figure 3: Temporal generalization study comparing the robustness of our proposed AIR, standard IB, and BCE loss. Subplots show performance via (a) BACC and (b) F1-score on the original data split versus a split with an enlarged temporal delay (denoted with "/T").

5.5 Temporal Generalization Study (RQ3)

To further verify the robustness of AIR module in handling temporal non-stationarity, we design a cross-period generalization study. This experiment enlarges the time gap between the training set and the test set, aiming to verify whether the AIR module can model temporal non-stationarity and maintain more stable performance when there exists a significant temporal span between training and testing data. Specifically, we use two AIR-related variants from RQ3 and conduct comparisons on two sets of datasets: (1) using the original data split (AIR, IB, and BCE); (2) keeping the test set unchanged, but training the models only with time-delayed data (AIR/T, IB/T, BCE/T). Additionally, to better demonstrate the robustness of the prediction results, we use box plots instead of mean values to show the distribution of the results for each prediction. The results are reported in Figure 3, which illustrates the distribution of the 60-day prediction results. It can be concluded that when faced with the challenge of an increased time span, the performance of the BCE and IB models decline significantly. This is particularly evident in the box plot for IB/T, which shows a substantially lower median performance and a highly scattered outlier distribution. In contrast, AIR/T exhibited only a slight decrease in median performance compared to AIR and maintains a low variance. This strongly demonstrates that AIR can learn more generalizable features, effectively resisting the performance decay caused by non-stationarity.

5.6 Case Study (RQ4)

To intuitively demonstrate our workflow in a real-world scenario, this section conducts an analysis of a specific case study. As illustrated in Figure 4, we trace the evolutionary path of an initial observed event, "Expressing Cooperative Intent" (Event Code: 03.0).

First, PER module is designed to identify the most significant precursor events from several candidates. As shown in Figure 4, the PER module evaluates multiple potential evolutionary processes using a multi-view entropy path routing mechanism. For instance, the path "03.0 -> Economic Cooperation" (06.1) is selected by the model, while the path "03.0 -> Halt Negotiations" (12.0) is discarded. This filtering process is the first step toward enhancing interpretability, as it clearly demonstrates the associations for path selections.

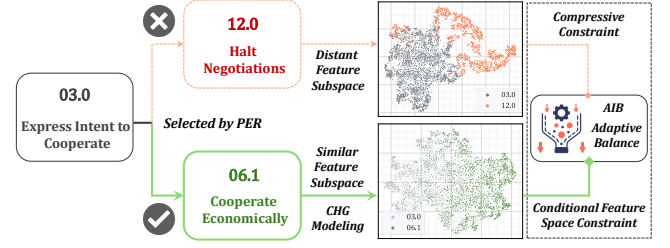


Figure 4: Case study of Mediator Learning, which shows an example event: "03.0: Express Intent to Cooperate".

After PER module, CHG module is proposed to model precursor events into the evolutionary process. The feature subspace visualization in Figure 4 shows that the feature representation of "Economic Cooperation" (06.1) is located within a similar feature subspace with the initial event "Expressing Cooperative Intent" (03.0). This indicates a higher degree of consistency between the two events in terms of temporal dynamics and event patterns, allowing the model to capture their correlated and continuous evolutionary relationship. In contrast, "Halt Negotiations" (12.0) resides in a Distant feature subspace, indicating that its feature pattern differs significantly from the current evolutionary path. Through this approach, CHG not only validates the effectiveness of the paths selected by the PER module, but also provides solid evidential support.

Finally, the AIR module plays the role of dynamic optimization and adaptive trade-off throughout the learning process. In this case, AIR achieves this objective through its unique constraint mechanisms: (1) Conditional Feature Space Constraint: this constraint guides the model to focus on the feature information extracted from the effective precursor event "Cooperative Economy" (06.1), thereby reinforcing the information flow within similar feature subspaces. (2) Compressive Constraint: this constraint actively suppresses noisy information from irrelevant paths such as "Negotiation Breakdown" (12.0), preventing it from interfering with the final prediction. This not only enhances the robustness of the model's predictions, but also makes its decision-making more transparent.

6 Conclusion

To address the challenges of extreme class imbalance and temporal non-stationarity in rare social event prediction, this paper introduces *Mediator Learning*. It models critical mediators, decomposing the non-trivial task into more manageable stages to better capture the evolutionary process. We propose three modules: a Precursor Event Router (PER) to identify precursor events, a Conditional Hierarchical Graph Network (CHG) to model the evolutionary process, and an Adaptive Information Regularizer (AIR) to model temporal non-stationarity. Experiments on large-scale, real-world datasets demonstrate that our model significantly outperforms baselines.

Acknowledgments

This work was supported by the National Natural Science Foundation of China under Grants (No.62532003, 62325206), and the Key Research and Development Program of Jiangsu Province under Grant BE2023016-4.

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A Notations

In this section, we introduce the notations used in this paper.

Table 4: Notations and Descriptions

Notations	Definitions and Descriptions
$X_{0:t}$	Time series input for all observed events ranging from time 0 to t
$M_{t:t+\Delta t}$	Probability distribution for mediators ranging from time t to $t + \Delta t$
$Y_{t+\Delta t}$	Occurrence probability for target rare events at time $t + \Delta t$
$I(\cdot; \cdot)$	Mutual information of two variables
T_h, T_s, T_o	The structural score, co-occurrence score and semantic score among different events.
$\mathbf{T}, T_{ij}$	Event transition matrix and the value at i -th row and j -th column
Path	Routed path in transition matrix $\mathbf{T}$ for target
$X_{0:t}^{\mathcal{R}}, X_{0:t}^{\mathcal{M}}, X_{0:t}^{\mathcal{E}}$	Observed time series for rare events, precursor events and all observed events.
H_*, A_*	$* \in (\mathcal{R}, \mathcal{M}, \mathcal{E})$ refers to the node embedding and adjacency matrix
(α, β)	Shape parameters of Beta-Bernoulli distribution
$\mathbf{E}_m, \hat{\mathbf{E}}_m$	Raw aggregated representation and conditional-gated aggregated representation
$\mathbb{E}_{p(X)}, \mathbb{E}_{p(M)}$	Approximated mathematical expectation for $p(X)$ and $p(M)$

B Baselines

We evaluate our approach against three categories of comprehensive baselines: (1) Time series classification methods, which capture temporal dependencies via diverse mechanisms. We include MiniROCKET [7] with random convolutional kernels, TSCMamba [1] using state space models, MTG [15] using temporal graphs, and Series2Vec [10]. GRU [5] is adopted as a standard recurrent baseline. (2) Imbalanced learning methods, which mitigate class disparity through data augmentation or representation shaping. Specifically, we compare against T-SMOTE [41] for temporal synthesis, and contrastive learning approaches like SmdCLR [19] and DGMSCL [31] that enhance separability. (3) Information Bottleneck methods, which optimize the trade-off between representation compression and predictive accuracy. We select GIB [38] and InsGNN [9] to evaluate the efficacy of IB principles in filtering noise from graph structures. All methods are trained using optimal parameters reported in their respective papers. For baselines without public code, we reproduce the architectures and report the best experimental results.

C Metrics and Implementation Details

Rare event prediction suffers from extreme class imbalance, the traditional accuracy metric may be misleading. Therefore, to comprehensively assess the model’s performance, we have adopted the following three key metrics:

$$\begin{aligned}
 \text{BACC} &= \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) \\
 \text{F1} &= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \\
 \text{MCC} &= \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}
 \end{aligned} \tag{13}$$

where TP is the number of samples correctly predicted as positive, TN is the number of samples correctly predicted as negative, FP is the number of samples incorrectly predicted as positive, FN is the number of samples incorrectly predicted as negative.

BACC is the arithmetic mean of the True Positive Rate (TPR) and the True Negative Rate (TNR). It treats each class equally, regardless of its sample size. The F1-Score is the harmonic mean of Precision and Recall. This metric primarily focuses on the model’s performance in predicting the positive class (i.e., the rare events we are interested in). MCC is considered one of the most robust and balanced metrics for evaluating binary classifiers on imbalanced datasets. It takes into account all four values in the confusion matrix (TP, TN, FP, FN) and its value ranges from -1 to +1.

All experiments in this paper are carried out on a Linux workstation equipped with an AMD EPYC 7542 32-Core CPU, an NVIDIA GeForce RTX 4090D GPU, and 512GB of RAM. The code is implemented in Python 3.10 with PyTorch 2.2.2 and CUDA 11.8. To achieve our best practice, we employ AdamW as optimizer with learning rate chosen from $[1e - 3, 5e - 4, 1e - 4]$. The dropout probability is set to 0.2 and weight decay is set to $1e - 2$. The model is trained for up to 100 epochs, with an early stopping strategy applied to prevent overfitting. Specifically, if the BACC metric on the validation set does not improve for 10 consecutive epochs, training will be terminated early. The dimensionality of node embeddings for all events, hidden layers, and the graph convolutional network is uniformly set to 64. When generating the event transition matrix, the time window size used to compute the co-occurrence score is set to 7 days. The semantic score is computed based on a pre-trained Sentence-BERT model (all-MiniLM-L6-v2) to extract embedding vectors from event description texts. For compressive constraint, we adopt a Masked Autoencoder with masking ratio 75% applied to the input time series. Our code is available at <https://github.com/JashinKorone/MediatorLearning>.

D Discussion on Event Transition Matrix

The event transition matrix $\mathbf{T}$ constitutes the core of PER module, where the structural score is introduced to capture hierarchical relationships among events from a taxonomic perspective. Specifically, we employ the Conflict and Mediation Event Observations (CAMEO) coding scheme, a standardized taxonomy widely adopted in political event analysis. As illustrated in Figure 5(a), CAMEO organizes events in a tree-structured hierarchy, in which each event

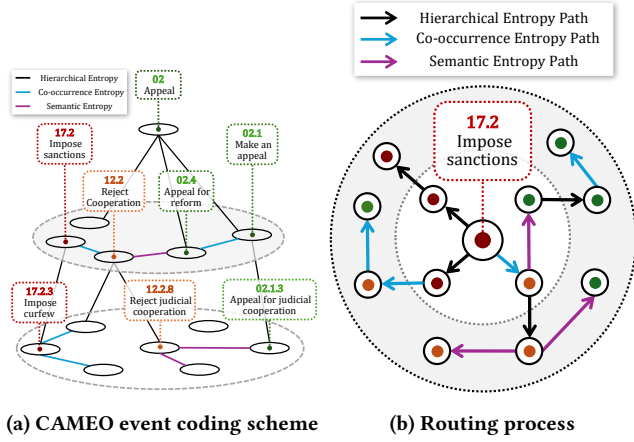


Figure 5: Details on event transition matrix

is assigned a unique numeric code. The structural score is computed as the shortest path distance between two events within the CAMEO hierarchy. Conceptually, we treat the CAMEO taxonomy as an undirected graph, where each code node is connected to its parent and child nodes. A shorter distance between two codes indicates stronger semantic relatedness and implies a higher prior probability of transformation between the corresponding events. For instance, the distance from “Impose Sanctions” (17.2) to “Impose Curfew” (17.2.3) equals 1, whereas the distance to “Make an Appeal” (02.4) is substantially larger. This taxonomy-informed distance metric provides structured prior knowledge that facilitates the model’s inference capacity.

E Discussion on CHG module

In the CHG module, we employ the Beta-Bernoulli distribution to estimate the occurrence probability of precursor events. This choice is motivated by the distribution’s unique ability to provide both probabilistic predictions and associated uncertainty estimates. Specifically, the application proceeds in two stages: first, the model samples a probability value p from a Beta distribution. Rather than being a fixed scalar, this value represents a distribution that reflects the model’s confidence. Subsequently, p is used as the parameter of a Bernoulli distribution to determine whether the event occurs. This hybrid formulation is critical because it not only outputs predictive probabilities in a manner similar to traditional classifiers but also quantifies the confidence of those predictions. Such uncertainty estimation is essential, as it mitigates error propagation and thereby enables more robust and reliable learning.

F Derivation and Discussion on Lemma 2

In this chapter, we first derive the formula presented in Lemma 2, and then proceed to prove its conformity with the principles of optimal compression theory. Firstly, Lemma 2 is an conditional extension derived by Lemma 1, which can be originally written as:

$$\min_{\phi, \theta} \underbrace{I_{\phi}(X_{0:t}; M_{t:t+\Delta t} | \mathbf{E}_m)}_{(a)-(b)} - \gamma \underbrace{I_{\theta}(M_{t:t+\Delta t}; Y_{t+\Delta t})}_{(c)} \quad (14)$$

According to the chain rule for mutual information $I(A; B, C) = I(A; C) + I(A; B|C)$, we can rewrite the first term in Eq.(14) into:

$$\underbrace{I_{\phi}(X_{0:t}; M_{t:t+\Delta t} | \mathbf{E}_m)}_{(a)-(b)} = \underbrace{I_{\phi}(X_{0:t}; M_{t:t+\Delta t}, \mathbf{E}_m)}_{(a)} - \underbrace{I_{\phi}(X_{0:t}; \mathbf{E}_m)}_{(b)} \quad (15)$$

which is equivalent to the format of AIR in Lemma 2. Furthermore, we can derive that:

$$\begin{aligned} \underbrace{I_{\phi}(X_{0:t}; M_{t:t+\Delta t} | \mathbf{E}_m)}_{(a)-(b)} &= \mathbb{E}_{p(X, M, \mathbf{E}_m)} \left[\log \frac{p(M|X, \mathbf{E}_m)}{p(M|\mathbf{E}_m)} \right] \\ &= \mathbb{E}_{p(X, \mathbf{E}_m)} \left[\int p(M|X, \mathbf{E}_m) \log \frac{p(M|X, \mathbf{E}_m)}{p(M|\mathbf{E}_m)} dM \right] \\ &= \mathbb{E}_{p(X, \mathbf{E}_m)} [D_{KL}(p(M|X, \mathbf{E}_m) || p(M|\mathbf{E}_m))] \end{aligned} \quad (16)$$

In practice, the true posterior distribution $p(X, M, \mathbf{E}_m)$ and prior distribution $p(M|X, \mathbf{E}_m)$ are unknown. Therefore, we introduce variational distributions, parameterized by ϕ , to approximate them. Through variational inference, this approach yields a tractable upper bound. For the implementation, we assume these approximate posterior and prior distributions are both Gaussian and directly use the KL divergence between them as the optimization objective. This is the conditional feature space constraint in $\mathcal{L}_{\text{AIR}}$.

Regarding the term (b), our goal is to maximize $I(X_{0:t}; \mathbf{E}_m)$. Since exact computation is intractable, we resort to maximizing its variational lower bound. By introducing a variational distribution $q_{\psi}(X_{0:t} | \mathbf{E}_m)$ (implemented as the decoder in the Masked Autoencoder), we have:

$$\begin{aligned} I(X_{0:t}; \mathbf{E}_m) &= H(X_{0:t}) - H(X_{0:t} | \mathbf{E}_m) \\ &= H(X_{0:t}) + \mathbb{E}_{p(X, \mathbf{E}_m)} [\log p(X_{0:t} | \mathbf{E}_m)] \\ &\geq \mathbb{E}_{p(X, \mathbf{E}_m)} [\log q_{\psi}(X_{0:t} | \mathbf{E}_m)] + \text{const} \end{aligned} \quad (17)$$

where $H(\cdot)$ denotes the entropy. Assuming q_{ψ} follows a Gaussian distribution, maximizing the log-likelihood is equivalent to minimizing the Mean Squared Error (MSE) between the input $X_{0:t}$ and the reconstruction $\hat{X}_{0:t}$. Thus, minimizing the reconstruction loss serves as a proxy for maximizing $I(X_{0:t}; \mathbf{E}_m)$.

Also, for the second term in Eq.(14), its objective is to maximize the mutual information $I_{\theta}(M_{t:t+\Delta t}; Y_{t+\Delta t})$, which is approximated by a lower bound on the negative log-likelihood. According to the definition, we can rewritten the mutual information into:

$$\begin{aligned} \underbrace{I_{\theta}(M_{t:t+\Delta t}; Y_{t+\Delta t})}_{(c)} &= \mathbb{E}_{p(M)} \left[\log \frac{p(Y|M)}{p(Y)} \right] \\ &= \mathbb{E}_{p(M)} [\log p(Y|M)] - \mathbb{E}_{p(Y)} [\log p(Y)] \\ &= \mathbb{E}_{p(M)} [\log p(Y|M)] + H(Y) \end{aligned} \quad (18)$$

where $H(Y)$ is the entropy of the target Y , which is a constant. Therefore, maximizing $I(M; Y)$ is equivalent to maximizing the term $\mathbb{E}_{p(M, Y)} [\log p(Y|M)]$. Maximizing this log-likelihood term is equivalent to minimizing its Negative Log-Likelihood (NLL). This forms the predictive constraint term of $\mathcal{L}_{\text{AIR}}$.

It is also worth noting that in Eq.(14), we only apply conditioning on bottom-level propagation. This is because the conditional constraint on upper-level propagation will **accidentally penalize the correlation between mediator and prediction results**, leading to violation of the optimal compression principle. If we apply conditional mechanism on the second term of information bottleneck in Eq.(14), our objective becomes: $\min_{\phi, \theta} I_{\phi}(X_{0:t}; M_{t:t+\Delta t} | \mathbf{E}_m) - \gamma I_{\theta}(M_{t:t+\Delta t}; Y_{t+\Delta t} | \mathbf{E}_m)$. According to the chain rule, the second term of our new objective can be rewritten as: $I_{\theta}(M_{t:t+\Delta t}; Y_{t+\Delta t} | \mathbf{E}_m) = I_{\theta}(M_{t+\Delta t}; Y_{t+\Delta t}, \mathbf{E}_m) - I_{\theta}(\mathbf{E}_m; Y_{t+\Delta t})$. It turns out that the first term $I_{\theta}(M_{t:t+\Delta t}; Y_{t+\Delta t}, \mathbf{E}_m)$ can be bounded by the same loss function as shown in *Lemma 2*, which suggests that this term encourages the conditional feature space learning. However, to maximize the new objective, we need to minimize $I_{\theta}(\mathbf{E}_m; Y_{t+\Delta t})$, which contradicts with the optimal compression principle. The term $I_{\theta}(\mathbf{E}_m; Y_{t+\Delta t})$ measures the mutual information for predicting the future $Y_{t+\Delta t}$. Minimizing this term may accidentally discarding the information in $\mathbf{E}_m$ that is useful for prediction.

G Ethical Announcement

This research is conducted with a firm commitment to ethical and academic standards. All data utilized in this study is exclusively sourced from the Global Database of Events, Language, and Tone (GDELT), a publicly accessible repository that compiles information solely from open-source news media across World Wide Web. We want to emphasize that our work involves no private, personal, or confidential information; all analysis is performed within the terms of use for this publicly available dataset. Our primary objective is to advance the understanding of rare social events for the academic community, with potential applications in areas such as risk assessment and humanitarian planning. We acknowledge that any model trained on historical news data may inherit existing biases, and its outputs should be interpreted as probabilistic inferences based on observed event patterns, not as definitive future certainties. We are mindful of the potential for misuse of this technology and explicitly oppose any unethical applications.